

Dropshipping Automation Project

End-to-End Pipeline on n8n — From Social Demand to After-Sales

Project Documentation: Context, Process Classification, Actors & Agents, Modules

1. Context

The dropshipping market is saturated with sellers who rely on guesswork and manual research — browsing AliExpress for hours, copying competitors, and reacting to trends too late. Meanwhile, real demand signals are visible every day on social media: thousands of comments asking “*where can I buy this?*”, complaining about unavailable products, or expressing desire for items seen in viral posts.

This project addresses that gap with an **end-to-end automated dropshipping pipeline built on n8n**. The system autonomously discovers products people already want by listening to social media chatter, validates them against supplier data for profitability, creates marketing content to generate the trend, and pushes the products directly into the seller's website (via its Git repository and database). The entire pipeline runs on a user-configured schedule (daily, weekly, or monthly) with notifications at every step — from discovery to after-sales.

Item	Description
Problem	Manual product research is slow, subjective, and reactive.
Solution	A scheduled, data-driven automation that turns social demand into a live store, hands-free.
Trigger	Cron schedule — user-configured: daily, weekly, or monthly.
Core technologies	n8n (orchestration), social & marketplace APIs/scraping, Git integration, database, AI content generation.

2. Classification of the Process

The system is classified as an **automated, event-scheduled, multi-stage business process** — specifically a **pipeline (chain) process**, where the output of each stage feeds the next, with decision points (scoring gates) between stages. In BPM terms, it is a scheduled orchestration workflow composed of ten functional modules, with error handling and notification as transversal concerns.

Dimension	Classification
Trigger type	Time-based (cron schedule — user-configured: daily / weekly / monthly).
Process structure	Sequential pipeline with feedback loops (after-sales data feeds back into scoring).
Automation level	Full-cycle automation with human-in-the-loop at the Git PR review gate.
Process nature	Discovery → Decision → Action → Support (information gathering, transformation, publication, service).
Execution mode	Batch — each run processes a batch of signals → shortlist → products.
Integration type	Cross-platform orchestration (social APIs, marketplaces, Git, database, messaging).

3. Acteurs et Agents

3.1 Human Actors

Actor	Role
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Store Owner (User)	Configures the schedule and settings (frequency, platforms, suppliers, products per run, notification channel); reviews Git pull requests before products go live; receives run reports and alerts.
Customers	Discover products through the generated social media posts/videos, buy on the website, and interact with after-sales support.

3.2 Software Agents (Automated)

Agent	Responsibility
Listening Agent	Scrapes X, Facebook, Instagram for demand signals.
Research Agent	Queries Amazon, Alibaba, AliExpress for sourcing data.
Scoring Agent	Ranks products by demand, margin, competition, and trend.
Content Agent	Generates and publishes social media posts and videos.
Publishing Agent	Syncs products to the website via Git commits and database writes.
Fulfillment Agent	Forwards orders to suppliers and tracks shipments.
Care Agent	Handles post-purchase emails, FAQ replies, and refunds.
Reporting Agent	Sends digests and alerts to the store owner after each run.

4. Modules

The pipeline is composed of ten functional modules. Each run follows the same sequence: **search** → **research** → **scoring** → **content** → **Git** → **database** → **notification**, with fulfillment and after-sales handling the post-purchase phase.

#	Module	Description
1	1. Demand Signals Scraper (Social Listening)	Monitors X, Facebook, and Instagram comments/posts for recurring "where can I buy this?" patterns. Turns raw social chatter into a ranked list of products people already want — before most sellers notice.
2	2. Trend Analysis & Product Scoring Engine	Aggregates scraped demand data, cross-references search/social velocity, and scores each product. Outputs a shortlist: high demand, low competition — no gut-feel guessing.
3	3. Supplier & Sourcing Finder	Matches each shortlisted product against Amazon, Alibaba, and AliExpress listings. Pulls price, reviews, shipping times, and minimum order quantities — so the system knows exactly where to source and at what margin.
4	4. Content & Trend Creation Engine	Generates social media posts and short videos for each validated product, then schedules and publishes them on the store's accounts. Creates the buzz that drives traffic to the store — automatically.
5	5. Store Sync via Git (Website Integration)	Connects to the sales website's Git repository and auto-generates product pages from the validated shortlist. Commits, pushes, and opens a PR — the store updates without human code changes.
6	6. Database Writer & Publish Log	Upserts trending products into the website's database and maintains a publish log. Nothing gets duplicated, and every run is traceable.
7	7. Order Forwarding & Fulfillment Bridge	Captures incoming store orders and forwards them directly to the supplier for processing. Pulls tracking numbers back and updates customers — hands-free from checkout to delivery.
8	8. After-Sales Care Module	Handles the post-purchase loop: automated order confirmations, shipping updates, AI-assisted replies to customer questions, and refund request processing based on supplier policy.
9	9. Scheduling & Pipeline Orchestration	Runs the entire pipeline on the user's chosen rhythm — daily, weekly, or monthly — via cron. One master workflow chains all modules together and passes data between them.
10	10. Notification & Reporting Layer	Sends the owner a digest after every run: products found, scores, items added, and any errors. Instant alerts when something needs attention.

4.1 Pipeline at a Glance

Stage	Module(s)	Output
Discover	1 – Social Listening	Ranked demand signals
Decide	2 – Scoring, 3 – Sourcing	Validated product shortlist
Promote	4 – Content Engine	Published posts & videos

Stage	Module(s)	Output
Publish	5 – Git Sync, 6 – Database	Live product pages + logged entries
Sell	7 – Fulfillment Bridge	Orders forwarded & tracked
Support	8 – After-Sales Care	Emails, replies, refunds handled
Control	9 – Orchestration, 10 – Notification	Scheduled runs + owner reports